2N7000

N-CHANNEL ENHANCEMENT MODE FIELD EFFECT TRANSISTOR

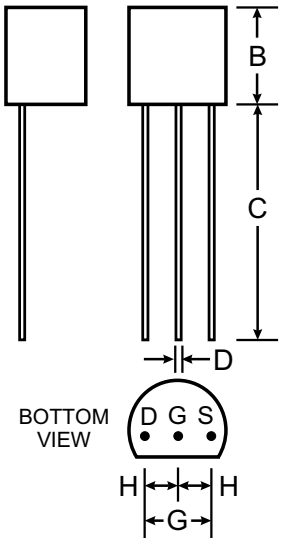
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Features

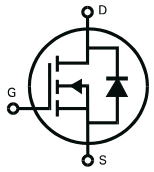
- Efficient High Density Cell Design Approaching 3 Million Cells per Square Inch
- Voltage Controlled Small Signal Switch
- Rugged
- High Saturation Current
- Low $R_{DS\ (ON)}$
- Fast Switching Speed

Mechanical Data

- Case: TO-92, Plastic
- Leads: Solderable per MIL-STD-202, Method 208
- Pin Connections: See Diagram
- Marking: Type Number
- Weight: 0.18 gram (approx.)



TO-92		
Dim	Min	Max
A	4.45	4.70
B	4.46	4.70
C	12.5	—
D	0.41	0.55
E	3.43	3.68
G	2.42	2.67
H	1.14	1.40
All Dimensions in mm		



Maximum Ratings @ T_A = 25°C unless otherwise specified

Characteristic	Symbol	Value	Units
Drain-Source Voltage	V_{DSS}	60	V
Drain-Gate Voltage $R_{GS} \leq 1.0M\Omega$	V_{DGR}	60	V
Gate-Source Voltage (pulsed)	V_{GSS}	±20	V
Drain Current Continuous Pulsed	I_D	115 800	mA
Total Power Dissipation Derating above T_A = 25°C	P_d	200 1.60	mW mW/°C
Maximum Terminal Temperature for Soldering Purposes	T_L	300	°C
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	625	K/W
Operating and Storage Temperature Range	T_j, T_{STG}	-55 to +150	°C

Notes: 1. Pulse width ≤ 300μs, duty cycle ≤ 2%.

Electrical Characteristics @ T_A = 25°C unless otherwise specified

Characteristics	Symbol	Min	Typ	Max	Unit	Test Condition
Off Characteristics						
Drain-Source Breakdown Voltage	$V_{(BR)DSS}$	60	—	—	V	V_{GS} = 0V, I_D = 10µA
Drain Cutoff Current	I_{DSS}	—	—	1.0	µA	V_{DS} = 48V, V_{GS} = 0V
Gate-Body Leakage Current	I_{GSS}	—	—	±10	nA	V_{GS} = ±15V, V_{DS} = 0V
On Characteristics (Note 1)						
Gate-Source Threshold Voltage	$V_{GS(th)}$	0.8	—	3.0	V	V_{DS} = V_{GS}, I_D = 1.0mA
Drain-Source On-Resistance	$r_{DS(ON)}$	—	—	5.0	Ω	V_{GS} = 10V, I_D = 0.5A
Dynamic Characteristics						
Input Capacitance	C_{ISS}	—	60	—	pF	V_{DS} = 25V, V_{GS} = 0V, f = 1.0MHz
Output Capacitance	C_{OSS}	—	25	—	pF	
Feedback Capacitance	C_{RSS}	—	5.0	—	pF	
Switching Characteristics						
Turn-On Time	t_{on}	—	10	—	ns	V_{DS} = 15V, I_D = 0.5A, V_{GS} = 10V, R_{GEN} = 25Ω, R_L = 25Ω
Turn-Off Time	t_{off}	—	10	—	ns	
Thermal Characteristics						
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	—	—	312.5	K/W	—

Notes: 1. Pulse width ≤ 300µs, duty cycle ≤ 2%.

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FOR NEW DESIGN
USE 2N7002**